AI-Powered Social Media Engagement Dashboard - Analysis File

1. Objective

The objective of this project is to analyze social media engagement data using an AI-powered dashboard to understand user interaction patterns, content performance, and platform effectiveness.

2. Dataset Overview

The dataset includes key social media metrics such as likes, comments, shares, impressions, reach, engagement rate, post type, and posting time.

3. Data Cleaning & Preprocessing

Data cleaning involved handling missing values, removing duplicates, standardizing date formats, and ensuring consistency in engagement metrics.

4. Key Metrics

Important KPIs include Engagement Rate, Total Likes, Total Comments, Total Shares, Impressions, and Reach. These metrics help evaluate content performance.

5. Exploratory Data Analysis

Analysis shows variation in engagement across different content types such as images, videos, and text posts. Video content generally shows higher engagement.

6. KPI Definitions

Engagement Rate = (Likes + Comments + Shares) / Impressions. Reach indicates the number of unique users who saw the content.

7. Observations

Posts published during peak hours show higher engagement. Consistent posting frequency improves audience interaction.

8. Business Recommendations

Focus on video-based content, optimize posting schedule, and leverage AI-based sentiment analysis to improve audience targeting.

9. Conclusion

The AI-powered dashboard provides actionable insights that help improve social media strategy and maximize engagement.